

PROTOCOL: DNA Extraction by Alkaline Lysis

v 1.0

*Alkaline lysis, also known as ‘HotShot’ (**Hot Sodium Hydroxide & Tris**), is a chemical DNA extraction method that is inexpensive and quick. It involves lysing tissue in an alkaline solution (pH 13) at hot temperatures and then neutralizing the reaction with an acidic solution (pH 5.5). The DNA can be purified after, but this is generally not necessary for PCR-based analyses.*

 *Bullet*  *points show tips for executing the protocol.*

(1) Prepare buffer solutions from concentrates

- a. Prepare Alkaline Lysis (AL) Buffer:
 - i. Centrifuge one 2 mL tube containing **1 mL** AL buffer concentrate.
 - ii. Pipette entire **1 mL** volume into included empty 50 mL tube with AL labels.
 - iii. Fill 50 mL AL tube to **50 mL** mark with molecular grade water.
 - iv. Invert to mix.
- b. Prepare Neutralization Buffer (NB)
 - i. Centrifuge one 2 mL tube containing **1 mL** NB buffer concentrate.
 - ii. Pipette entire **1 mL** volume into included empty 50 mL tube with NB labels.
 - iii. Fill 50 mL NB tube to **50 mL** mark with molecular grade water.
 - iv. Invert to mix.

Prepared buffer solutions can be stored at room temperature.

(2) Sample tissue into ONTOLOGY™ DNA extraction plates

-  *So long as the tips do not contact the wells, the same tips can be reused.*
-  *Pour the buffer solutions into a sterile reservoir and use a multi-channel pipette to dispense.*
-  *Ensure you label the DNA plates clearly from 1–10, because they come unlabeled*

- a. Fill the wells of the clear (transparent) ONTOLOGY™ DNA extraction plates with **50 µL** of **either** prepared AL buffer **OR** 95% EtOH (if using ethanol, you can reduce volume to **20 µL**).
 - i. If you will extract DNA today, sample tissue into lysis buffer.
 - ii. If you will extract DNA later, sample tissue into EtOH.
- b. Place a very small piece of tissue—at most **2 mm²** (half of a grain of rice)—into each well.
- c. Seal plate.
 - i. If you used AL buffer, seal with a **PCR seal**.
 - ii. If you used ethanol, seal with a **temporary seal**.
- d. Briefly centrifuge the plate to ensure that the tissue is submerged.



(3) Alkaline Lysis protocol

-  *If specimens are already in AL buffer, start from step **b**.*
-  *If using the droppers, note that these dispense approximately **40 µL**; use the droppers for both AL and NB to ensure the volumes are comparable.*
- a. Evaporate EtOH, if used, then add AL buffer (proceed to (b) if AL buffer already used):
 - i. Place plates containing EtOH uncovered (i.e., remove seal) in an incubator at **56°C**.
 - ii. Keep plates in incubator until all EtOH is evaporated from all wells, typically **30–45 minutes**, depending on the volume.
 - iii. Add **50 µL** AL buffer to each well after EtOH is evaporated.
 - iv. Apply **PCR seal** to plate.
- b. Quickly centrifuge the plate to ensure tissue is in contact with the AL buffer.
- c. Incubate sealed plate at **95°C** for **10 minutes**.
- d. After incubation, allow sealed plate to cool to room temperature for **3–4 minutes**.
- e. Centrifuge the sealed plate to remove any condensation that formed on the seal.
- f. Add **50 µL** of neutralization buffer using a pipette or the NB dropper if you previously used the AL dropper.
- g. Mix by gently tapping the plate.
- h. Store neutralized lysate at **–20°C** for long-term storage, **4 °C** for short term storage, or use immediately in PCR. If desired, tissue can be recovered at this point.

(4) (Optional) Tissue Recovery

- a. If carrying out tissue recovery e.g. for small organisms where the entire specimen was used for lysis, transfer the **full volume** (e.g. 100 µL) of neutralized lysate to a **new 96-well plate**, carefully avoiding the specimen.
- b. Add **50–100 µL** of **95% ethanol** to each original well containing the specimens to be retained. Cover the plate using a **foil seal** (or heat sealer, if available) and store at **–20°C** or store the vouchers as desired.



Preparation of stock reagents:

1. 1M Tris-HCl, pH 5.15
 - a. Dissolve **78.8 g** of Tris-HCl in **~450 mL** of deionized water (ideally filtered to 18.2 MΩ·cm).
 - b. Using 5M KOH, adjust pH to **5.15** (10–20 drops should suffice).
 - c. Adjust volume to **500 mL** with deionized/filtered water.
 - d. Filter sterilize solution through a 0.22 µm filter into a sterile container.
2. 0.5M EDTA, pH 8.0
 - a. Dissolve **9.31 g** of EDTA in **~40 mL** of deionized water (ideally filtered to 18.2 MΩ·cm).
 - b. Using NaOH pellets, adjust pH to **8.0**. Add one pellet at a time. EDTA will fully dissolve once pH hits 8.0.
 - c. Adjust volume to **50 mL** with deionized/filtered water.
 - d. Filter sterilize solution through a 0.22 µm filter into a sterile container.
3. 1M KOH
 - a. Dissolve **28.06 g** of KOH in **~450 mL** of deionized water (ideally filtered to 18.2 MΩ·cm).
 - b. Adjust volume to **500 mL** with deionized/filtered water.
 - c. Filter sterilize solution through a 0.22 µm filter into a sterile container.

Preparation of working solutions:

NOTE: To simplify preparation, the following two solutions are prepared directly in unopened 500 mL bottles of commercial molecular-grade water.

LYSIS BUFFER

Stock	Working Conc.	Volume (µL)
1M KOH	25 mM	12,828
0.5M EDTA pH 8.0	0.2 mM	205
100% Triton X-100	0.2%	103
Water	N/A	500,000
TOTAL	N/A	513,136

NEUTRALIZATION BUFFER

Stock	Working Conc.	Volume (µL)
1M Tris-HCl pH 5.15	40 mM	20,833
Water	N/A	500,000
TOTAL	N/A	520,833

Optional pH validation: If desired, measure the pH of lysis buffer (should be ~13), neutralization buffer should be between 5.5–6.0, and a 1:1 (v:v) mixture of both solutions (should be ~8.5).

